Department of Electrical and Computer Engineering
COMSATS University Islamabad, Attock Campus

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<tr><th colspan="3">Lab Rubrics Form</th></tr>
<tr><td>Lab Name</td><td colspan="2">Introduction to C Programming (Basic Input/Output and Variables)</td></tr>
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<tr><td colspan="3">Submitted by:</td></tr>
<tr><td colspan="2">Student Names</td><td>Registration #</td></tr>
<tr><td colspan="2">Sumaira Safeer</td><td>FA22-BCE-019</td></tr>
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Modern Tools Usage</td><td>Ability to execute the experiment utilizing hardware, software tools, or a combination of both.</td><td></td><td></td><td></td></tr>
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Communication Skills</td><td>Present lab activities effectively through detailed lab reports, oral examinations, and engaging presentations.</td><td></td><td></td><td></td></tr>
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Lab 01: Introduction to C Programming (Basic Input/Output and Variables)

Objective

To learn the basics of the C programming language, including writing simple programs, displaying output using printf(), declaring variables, performing arithmetic operations, and calculating the area of basic geometric shapes.

Software Requirements

- Code::Blocks IDE / Dev-C++ / Visual Studio Code .
- GCC Compiler.
- Windows Operating System.

In Lab Tasks:

1. Print "Hello World".
2. Print the student's name.
3. Display name, program, semester, grade, and GPA.
4. Input two values and display their sum.

Learning Outcomes

After completing this lab, the student will be able to:

- Understand the structure of a C program.
- Use header files and the main() function.
- Display output using printf().
- Declare and initialize variables.
- Perform basic arithmetic operations.
- Compile and execute C programs successfully.

Task#1:

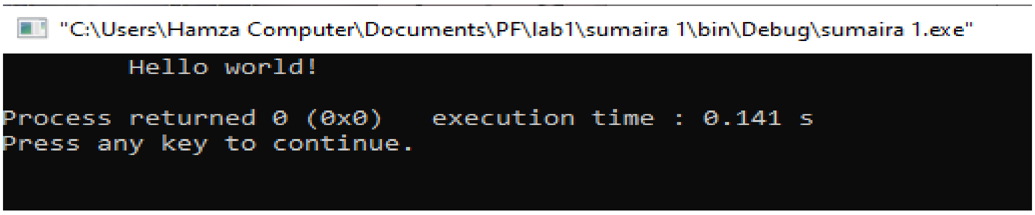
Write a program to print hello world.

Code:

```
#include <stdio.h>
 #include <stdlib.h>
 int main()
 {
    printf("\tHello world!\n");
    return 0;
```

```
}
```

Output:



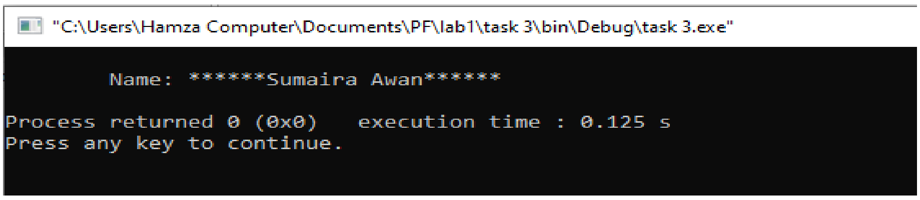
Task#2:

Write a program to write your name.

Code:

```
#include <stdio.h>
#include <stdlib.h>
int main()
{
   char Name[]= "******Sumaira Awan******";
   printf("\n\tName: %s\n",Name);
   return 0;
 }
```

Output:



Task#3:

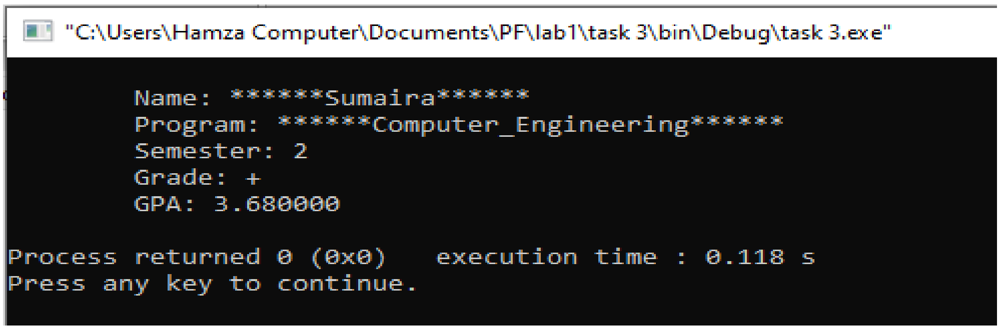
Write a program to print your name, semester, grade, GPA.

Code:

```
 #include <stdio.h>
#include <stdlib.h>
int main()
 {
```

```
 char Name[]= "******Sumaira******";
 char Sumaira_Program[]= "******Computer_Engineering******";
 int Sumaira_Semester= 2;
 char Sumaira_Grade= 'A+';
 float Sumaira_GPA= 3.68;
 printf("\n\tName: %s\n",Name);
 printf("\tProgram: %s\n",Sumaira_Program);
 printf("\tSemester: %d\n",Sumaira_Semester);
 printf("\tGrade: %c\n",Sumaira_Grade);
 printf("\tGPA: %f\n",Sumaira_GPA);
 return 0;
}
```

Output:



Task#4:

Write a program that input two values and output the sum of these values.

Code:

```
#include <stdio.h>
#include <stdlib.h>

int main()
{
  int P=6;
 int Q=2;
  int S=P+Q;
  /*sum of these vales*/
  printf("\n\t%d+%d=%d\n\t",P,Q,S);

   return 0;
```

```
}
```

Output:

```
"C:\Users\Hamza Computer\Documents\PF\lab1\task 3\bin\Debug\task 3.exe"

        6+2=8

Process returned 0 (0x0)   execution time : 0.141 s
Press any key to continue.
```

Post Lab Task:

1. Print a formatted pattern containing the student's name.
2. Calculate the area of a triangle and rectangle using given formulas.

Task#1:

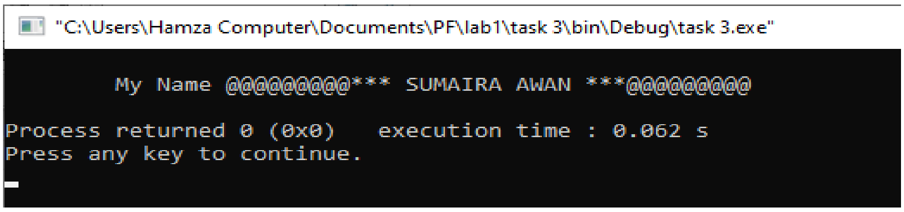
Write a Program that print
this **@@@@@@@@@@**
@@@@@@@@@@
*****Your**
Name***
@@@@@@@@
@@
@@@@@@@@
@@ ……….

Code:

```
#include <stdio.h>
#include <stdlib.h>

int main()
{
  char Name[]= "@@@@@@@@@@*** SUMAIRA AWAN ***@@@@@@@@@@";
  printf("\n\tMy Name %s\n",Name);
  return 0;
}
```

Output:



Task#2:

Write a program that calculate the area of Triangle and Rectangle by taking the values length and width as an input, where formulas are given below.

Area of triangle = length ∗
width Area of
triangle=2(lengt+width)

Code:

```
#include <stdio.h>
#include <stdlib.h>

int main()
{
  int Length= 07;
   int Width= 18;
   int Area_of_Tri,
Area_of_Rec;

Area_of_Tri=(Length*Width);

Area_of_Rec=2*(Length+Widt
h);

   /* Area of Triangle */
   printf("\nArea of Triangle =
       %d meter sq.",
Area_of_Tri);

   /* Area of Rectangle */
   printf("\nArea of Rectangle =
```

```
        %d sq.units\n",
Area_of_Rec);
  return 0;
}
```

Output: